

Daniel (Dan) Ruelas-Petrisko

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SUMMARY

I am a full-stack computer engineer with broad experience developing ASIC flows; from architectural specification to RTL, physical design, and EDA tooling. My research explores leveraging open-source tools and IP to minimize iteration time, without sacrificing design quality. I have experience leading multiple tapeouts in TSMC 28nm and GlobalFoundries 12nm, as well as prototyping designs in Skywater 130nm, TSMC 40nm and advanced FPGA prototypes.

EDUCATION

Ph.D., M.S. **University of Washington, Seattle** Computer Science & Engineering (2025, 2023)
A Qualitative Approach to Agile Hardware Design, advised by Professors Michael Taylor, Mark Oskin
M.S., B.S. **University of Illinois at Urbana-Champaign** Electrical & Computer Engineering (2018, 2016)
Architectural Exploration of Si-IF Many-Die Processors, advised by Professor Rakesh Kumar

EXPERIENCE

Organization Director Nov 2022 - present
Free and Open-Source Silicon (FOSSi) Foundation London, UK

- Advocates and provides stewardship for various open-source silicon tools, standards, and IP
- Organizes premiere conferences in the open-source silicon community, including LatchUp and ORConf

ASIC Design Consultant Jan 2023 - present
OpenASIC Consulting, LLC Seattle, WA

- Provides full-stack support for specification, implementation and verification across RTL2GDS flow
- Specializes in open-source IP integration, commercial/open hybrid CAD flows and rapid-design, N=1 test chips

Research Assistant Sep 2018 - August 2025
University of Washington Seattle, WA

- Collaborated on agile tapeouts of complex, record-breaking 12nm chips, hacking from RTL through GDS
- Created and maintained plethora of IP and tooling across the open-source BaseJump ASIC development ecosystem
- Mentored 20+ masters and undergrad projects, resulting in dozens of contributions to major research projects

Lead Digital Chip Architect - MaxSDR ASIC Mar 2021 - Dec 2022
MaXentric Technologies, LLC San Diego, CA

- Architected 3 Mgate TSMC 28nm mixed signal design with 192-core manycore and 3 Linux-capable cores
- Oversaw full RTL2GDS flow, including RTL, synthesis, floorplanning, verification, GLS and ECOs
- Pioneered source-synchronous link topology for rapid convergence of blocks, reducing iteration from 1 week to 1 day

Research Assistant August 2014 - May 2018
University of Illinois Champaign, IL

- Developed novel large-scale parallel simulation framework to approximate waferscale GPU system performance
- Applied novel ILP-based PPA optimization framework for multi-system architectural DSE of chiplets
- Implemented classical and novel ECC schemes in RTL to evaluate EDP tradeoffs for subthreshold operation

Logic Design Co-op Spring 2018; Summer 2018
IBM Austin, TX

- Designed and implemented VHDL for the POWER10 fetch unit, including branch prediction and issue logic
- Triaged core-level functional failures and power regressions alongside verification and integration teams

SoC Design Verification Intern Summer 2017
Apple Cupertino, CA

- Developed testbenches and verification flows for large-scale gate-level simulations, reducing build and run times
- Analysed coverage trade-offs between emulation and simulation flows, balancing developer and iteration time

Device Design Intern; Systems Validation Intern Summer 2016; Summer 2015
Intel Portland, OR

- Converged static timing and electrical quality for structurally designed circuits in embedded DRAM circuits
- Triaged hardware failures on prototype processors in lab, suggesting fix or debug directions to the design team

Embedded Software Intern; Test Automation Intern Summer 2014; Summer 2013
John Deere Des Moines, IA

- Automated embedded display testing equipment, eliminating costly manual programming and verification
- Created continuous build, test and deployment pipelines for production webapps

SKILLS

Hardware	Synopsys DC, ICC2, Primetime; Cadence Genus, Innovus, Tempus; Yosys, Verilator, Surelog
Software	SystemVerilog, C, C++, Python, TCL, Make, Git
Other	Logic Design, VLSI, Continuous Integration (CI), Agile Development, Architectural Simulation

PROJECTS

BlackParrot (Lead Architect)	 black-parrot <i>Heterogeneously tiled application-class multicore with custom cache coherence for accelerator integration</i>
ZynqParrot (Lead Architect)	 zynq-parrot <i>Cycle-accurate, Scale-Down, FPGA-accelerated co-emulation platform for generic RTL prototyping</i>
BSG Pearls (Lead Architect)	 bsg-pearls <i>Modular RTL and collateral library for hierarchy-insensitive hardware design</i>
HammerBlade (Maintainer)	 bsg_manycore <i>Highly scalable RISC-V manycore fabric for bulk-synchronous parallel workloads</i>
BaseJump STL (Maintainer)	 basejump_stl <i>Standard Template Library for latency-insensitive hardware design</i>

PUBLICATIONS

“A 12nm FinFET Implementation of Ruche Networks on a 99mm ² 2048-Core HammerBlade RISC-V Manycore SoC”	
P. Gao, D. Jung, S. Davidson, D. Ruelas-Petrisko , et al.	<i>TVLSI 2025*</i>
“A Qualitative Approach to Agile Hardware Design”	
D. Ruelas-Petrisko	<i>PhD Dissertation, ProQuest 2025*</i>
“ZynqParrot: A Scale-Down Approach to Cycle-Accurate, FPGA-Accelerated Co-Emulation”	
D. Ruelas-Petrisko , et al.	<i>arXiv 2025*</i>
“Scalable, Programmable and Dense: Open-Source RISC-V Manycore with Scalable Resource Organization”	
D. Jung, M. Ruttenberg, P. Gao, S. Davidson, D. Ruelas-Petrisko , et al.	<i>ISCA 2024</i>
“Scaling Program Synthesis Based Technology Mapping with Equality Saturation”	
G. Smith, C. Knizek, D. Ruelas-Petrisko , et al.	<i>WOSET 2024</i>
“RISE: RISC-V SoC for En/decryption Acceleration on the Edge for Homomorphic Encryption”	
Z. Azad, G. Yang, R. Agrawal, D. Ruelas-Petrisko , et al.	<i>IEEE TVLSI</i>
“RACE: RISC-V SoC for En/decryption Acceleration on the Edge for Homomorphic Computation”	
Z. Azad, G. Yang, R. Agrawal, D. Ruelas-Petrisko , et al.	<i>ISLPED 2022</i>
“BlackParrot: An Agile Open Source RISC-V Multicore for Accelerator SoCs”	
D. Ruelas-Petrisko , et al.	<i>IEEE Micro July/August 2020</i>
“NoC Symbiosis (Invited Paper)”	
D. Ruelas-Petrisko , et al.	<i>NOCS 2020</i>
“Architectural Exploration of Si-IF Many-die Processors”	
D. Ruelas-Petrisko	<i>MS Dissertation, ProQuest 2018</i>
“Design Space Exploration for Chiplet-Assembly-Based Processors”	
S. Pal, D. Ruelas-Petrisko , et al.	<i>IEEE VLSI April 2020</i>
“Rethinking Waferscale Processors - A GPU Case Study”	
S. Pal, D. Ruelas-Petrisko , et al.	<i>HPCA 2019</i>
“A Case for Packageless Processors”	
S. Pal, D. Ruelas-Petrisko , et al.	<i>HPCA 2018</i>
“Transforming Error Patterns to Enable Deeper Voltage Scaling in On-chip Memories”	
H. Duwe, X. Jian, D. Ruelas-Petrisko , et al.	<i>ISCA 2016</i>

WORKSHOP & TUTORIALS

FOSSi Latchup 2025	Introduced BSG Pearls as a modular library for hierarchy-insensitive hardware design
FOSSi LatchUp 2022	Introduced BSG Tag as an flexible, robust and minimal ASIC configuration bus
RISC-V Summit 2020	Tutorial to get up and running with BlackParrot and accelerators
Supercomputing 2020	Described integration BlackParrot into HPC systems
FOSDEM 2020	Announced the BlackParrot processor to the open-source hardware community
FOSSi LatchUp 2019	Pitched BaseJump STL as standard library for latency-insensitive hardware design

CHIP GALLERY

BP0	4-core BlackParrot	GF12 3x3mm
BP1	4-core BlackParrot	TSMC40 4x6mm*
BigBlade	16 x (2 uncore + CGRA + 128-core manycore)	GF12 10x10mm
MaxSDRv1	1 big, 2 little cores, 192-core manycore, high-speed I/O	TSMC28 4x6mm
TT-DLL	All-digital delay-locked-loop	Sky130 160x100um
MaxSDRv2	1 big, 2 little cores, 192-core manycore, high-speed I/O, LPDDR1	TSMC28 4x6mm
MiniBlade	DBI testing infrastructure, scalable clock generator	GF12 1x1mm*